Make Some Noise: Unsupervised Remote Sensing Change Detection Using Latent-Space Perturbations

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***Abstract*—Unsupervised change detection (UCD) aims to localize changes between two images of the same region without relying on labeled training data. Most recent approaches either use a frozen foundation model in a training-free manner or train with synthetic changes generated in pixel space. Both strategies inherently rely on predefined assumptions about change types, typically introduced through handcrafted rules, external datasets, or auxiliary generative models. Due to these assumptions, such methods fail to generalize beyond a few change types, limiting their real-world usage, especially in rare or complex scenarios. To address this, we propose make some noise (MaSoN), an end-to-end UCD framework that synthesizes diverse changes directly in the latent feature space during training. It generates changes dynamically estimated from feature statistics of the target data, enabling diverse yet data-driven variation aligned with the target domain. Since synthesis happens in latent space, it also easily extends to new modalities, such as SAR and multispectral data. MaSoN generalizes strongly across diverse change types and improves the average *F*1-score across five benchmarks by 14.1 percentage points (p.p.). Project page: https://blaz-r.github.io/mason_ucd/**

***Index Terms*—Change detection (CD), remote sensing, unsupervised learning.**

I. Introduction

CHANGE detection (CD) is a fundamental task in remote sensing that involves localizing changes in a sequence of images (typically two) of the same geographic region [1], [2], [3], [4]. This task is essential to various applications, including disaster response, urban development monitoring, and land use mapping [1], [3], [5], [6]. Recent supervised methods have demonstrated impressive results [1], [2], [7], [8], but they rely on pixel-level annotations and fail to generalize to unseen change types or new geographic contexts [9], [10], [11], [12].

Dependence on labels presents a critical limitation for time-sensitive applications and situations where annotation is unfeasible. In these situations, unsupervised change detection (UCD) offers an effective alternative [3], [4], [9], as it removes the need for manual labeling and enables the use of large archives of unlabelled bi-temporal imagery. By enabling fast analysis without human supervision, UCD methods support rapid response [10], [12], [13], [14] and efficient large-area monitoring with reduced reliance on labor-intensive annotation pipelines [3].

Recent UCD methods largely follow two strategies, each with limitations. *Training free approaches* [4], [15], [16], [17] leverage frozen priors in a training-free manner (e.g., segment anything model (SAM) [18]), but these are typically derived from natural/urban imagery and degrade under domain shift [10] (e.g., landslides or croplands). State-of-the-art *learning-based methods* aim to overcome this by adopting a two-stage pipeline: they first *generate* synthetic changes in pixel space and then *train* a change detector on the generated samples [19], [20], [21], [22]. However, the generated changes are shaped by assumptions introduced via auxiliary models [21], handcrafted rules [20], [23], or external datasets [19], [24], [25], which limits diversity and generalization to novel change types. Moreover, methods from both families are in most cases restricted to RGB-only input and often struggle with seasonal or radiometric variations (which we refer to as *irrelevant changes).*[1]

We address these limitations with make some noise (MaSoN), a novel end-to-end UCD framework that trains on dynamically generated changes in the latent space. Instead of pixel-space augmentations, MaSoN injects Gaussian noise into the feature maps of a pretrained encoder. The noise scale is dynamically estimated using feature statistics from the target data, enabling on-the-fly generation of decoupled relevant and irrelevant changes. This design introduces diverse domain-data-aligned training signals without any labeled external data or a multistage setup. Because change synthesis occurs in feature space, MaSoN is modality-agnostic and can be extended to new modalities such as multispectral and SAR with a simple encoder swap. As a result, MaSoN achieves superior performance across different change types and consistently outperforms prior methods across multiple datasets, as illustrated in Fig. 1.

Our contributions are twofold as follows.

1) We propose the first end-to-end latent-space change generation and detection framework that can be trained in an

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[1]Seasonal changes can be relevant in some applications, but we follow the majority of related work and benchmarks that consider them irrelevant.

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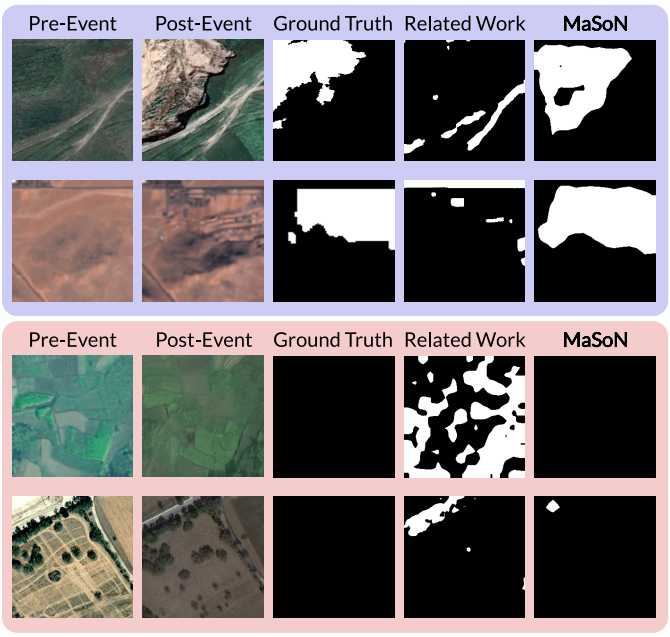


Fig. 1. Related methods that rely on foundation models in a training-free manner or on changes generated in pixel space often produce noisy or inaccurate masks, either missing relevant changes or overreacting to irrelevant changes. By generating changes in latent space, MaSoN learns from diverse and more data-aligned variations, producing better predictions with fewer false positives. This highlights MaSoN's improved ability to generalize across diverse and challenging change scenarios.

unsupervised manner with our on-the-fly change synthesis strategy. Our framework addresses the change-type generalization problems in related approaches, thereby improving CD performance across diverse change types.

2) We introduce a procedure for creating synthetic changes inside the latent space of an encoder. The synthetic changes are modeled as Gaussian noise, which is dynamically estimated using simple statistics derived from the latent features of the target data. Through experiments and theoretical analysis, we show that real-world changes can be decoupled into irrelevant and relevant categories and approximated accordingly. This strategy enables variability that is hard to capture in pixel space, and it naturally supports non-RGB modalities if a suitable encoder exists.

We evaluate MaSoN on five benchmark datasets spanning diverse change types, including natural disasters, building changes, urban development, and cropland changes. MaSoN achieves superior performance on four out of five datasets, outperforming prior methods by an average of 14.1 percentage points (p.p.), a relative increase of 38.6%. We also demonstrate that MaSoN *extends* and achieves strong results on multispectral and SAR modalities, which most related methods cannot easily do.

II. Related Works

Remote sensing CD is a long-established task, with early approaches based on pixel-level comparisons and statistical techniques [26], [27], [28], [29]. Deep learning advancements have since enabled end-to-end CD using Siamese convolutional architectures [1], [7], [30] which were later extended to transformers [2], [31], [32], [33], state-space models like Mamba [8], [34], [35] and diffusion-based approaches [19], [24], [36]. However, most recent models remain supervised and heavily rely on scarce annotated datasets, which are difficult to collect, especially for rare events such as natural disasters [3], [10]. Furthermore, such models often fail to generalize to unseen scenarios [9], [10], requiring costly supervised retraining.

Unsupervised CD addresses these limitations by operating solely on unannotated image pairs, making it especially valuable for rapid disaster response and other time-critical applications where annotated data is unavailable [4], [9], [10]. Methods like image-differencing [26], Markov random fields [37], and change vector analysis (CVA) [37], [38] rely on probability theory and statistics. While historically important, they are limited in expressiveness and are outperformed by modern deep-learning counterparts [4], [15], [16], [39].

Another branch of unsupervised CD additionally considers image pairs acquired by different sensors (heterogeneous CD). IRG-McS [40] represents the images as graphs and detects changes through cross-image graph mappings, whereas RIEM [41] infers changes through a rules-induced energy model. These methods explicitly mitigate modality differences through designed structural or relational constraints.

Several recent UCD approaches leverage large vision foundation models (VFMs). AnyChange [4], SCM [42], and DynamicEarth [17] use SAM [18] for segmentation, and CDRL-SA [43] uses it for refinement. Although effective in some scenarios, these models rely on SAM with frozen prior knowledge from the natural image domain, making them less suitable for remote sensing tasks. In contrast, our method is learning-based, allowing representations to adapt to the domain, thus improving generalization and flexibility.

Other learning-based methods create synthetic changes in pixel-space [20], [21], [24], [44]. ChangeStar [25] constructs synthetic data from unrelated segmentation datasets. Other methods, such as CDRL [21], FCD-GAN [45], SyntheWorld [44], Changen [22], [24], and HySCDG [19] use generative models for style transfer or full image synthesis. These models require auxiliary external datasets to generate synthetic changes, which limits diversity and domain generalization. I3PE [20] and S2C [23] remove such dependencies but still operate in pixel space using handcrafted transformations, leading to constrained variability. In contrast, MaSoN generates synthetic changes directly in the latent feature space during training. These changes are dynamically estimated using feature statistics of the target data, enabling diverse yet domain-consistent changes and stronger generalization.

Learning with noise is an established strategy in deep learning for improving robustness and generalization. DN-DETR [46] uses query denoising to accelerate training and stabilize bipartite matching for *object detection*. In *anomaly detection*, noise was applied to simulate anomalies [47], [48], [49]. GeoCLIP [50] leverages noise to model GPS inaccuracies and for contrastive learning in *geolocalisation*. It has also been applied to mitigate label uncertainty in hyperspectral CD

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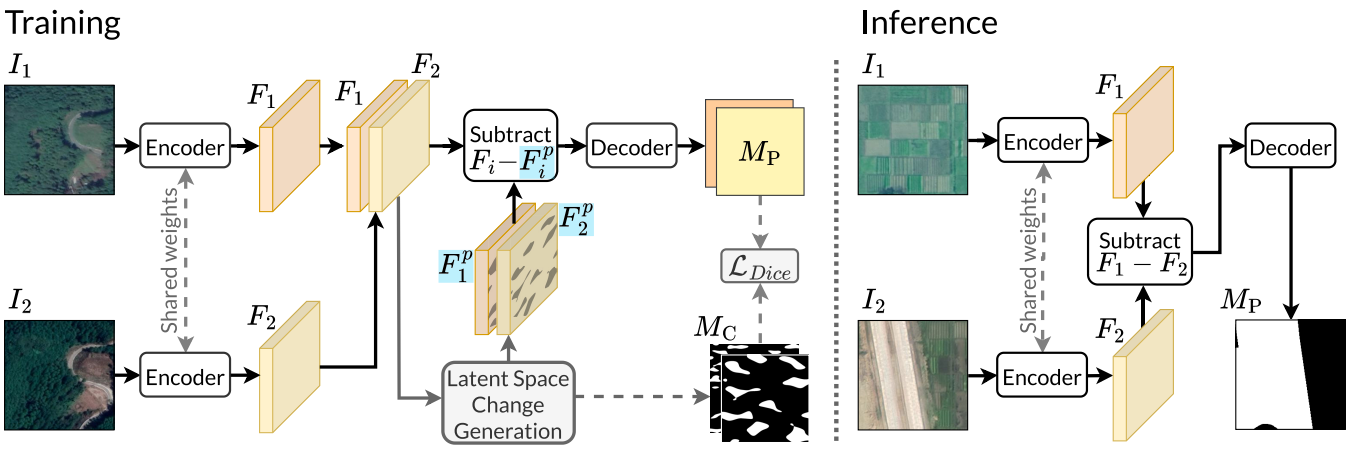


Fig. 2. Architecture of the proposed unsupervised CD framework, MaSoN.

[51]. Building on these foundations, MaSoN introduces a novel strategy to generate changes with Gaussian noise directly in the feature space. Unlike previous methods that rely on fixed [47], [52] and nondecoupled [53] Gaussian noise, MaSoN decouples it into irrelevant and relevant noise and dynamically adjusts it during training to address the inherent variability in remote sensing CD.

III. Let Us Make Some Noise

The proposed method (MaSoN) consists of a *shared weight encoder*, a *latent-space change generation strategy*, and a *mask decoder*. The features are first extracted using the *encoder* (Φ). Then, the *latent-space change generation strategy* is used (only during training) to generate synthetic changes on the feature level. Features from different time steps with generated synthetic changes are then fused (in our case via elementwise subtraction), and fed into the *mask decoder* ($\mathbb{D}$), which outputs the predicted change mask. The architecture of our method is depicted in Fig. 2 and explained in Sections III-A–III-E.

A. Encoder

Given a bi-temporal input image pair (I_1, I_2), we extract features (F_1, F_2) using a shared-weight pretrained encoder Φ. For each image I_i, the encoder produces a hierarchical feature set

$$\Phi(I_i) = F_i = \left\{ f_i^{(l)} \mid l \in L \right\} \tag{1}$$

where l denotes the layer level, and $f_i^{(l)}$ denotes the feature map at layer l from image I_i.

B. Why Gaussian Noise?

To better understand and motivate our synthetic change generation procedure, we analyze the behavior of features extracted by the pretrained encoder. Our goal is to determine whether changes can be meaningfully characterized and thus approximated directly in the latent space.

We perform the analysis on five different benchmark datasets (described later in Section IV). For each image pair, we extract features and compute elementwise differences at each layer: $f_{diff}^{(l)} = f_1^{(l)} - f_2^{(l)}$. Using the ground-truth annotations, we group these differences into changed and unchanged regions. Fig. 3 shows histograms of these difference values averaged across all datasets for each layer in the feature hierarchy.

1) Main Takeaways: According to our analysis, changed and unchanged regions exhibit distinct patterns in feature space. Unchanged areas produce small differences, resulting in sharp peaks near zero with a narrow distribution. Changed regions show a broader, heavier-tailed distribution, with fewer near-zero differences, especially in deeper layers that capture high-level semantic concepts. Although both distributions overlap and are *centered at zero*, their *variances differ*. From a theoretical perspective, the *maximum-entropy* principle states that, among all distributions consistent with the known constraints, the one with the largest entropy is the best representation of our current knowledge [54]. Given the observed zero mean and per-channel variances, the *Gaussian distribution satisfies these constraints* and *maximizes differential entropy* [55].

The theoretical grounding and empirical similarity to the Gaussian distribution suggest that both unchanged and changed regions can be synthetically generated (or rather locally approximated) during training using properly calibrated Gaussian noise. We later demonstrate that calibration can be achieved using batch statistics in order to adapt to diverse remote sensing scenes. An extended theoretical justification is provided in Supplementary Section E, while full empirical implementation details, per-dataset results, and discussion are given in Supplementary Section C.

C. Latent-Space Change Generation Strategy

As illustrated in Fig. 4, given feature map sets of an image pair, F_1 and F_2, our goal is to synthesize training pairs with controllable changes. Since image pairs may already contain (unknown) changes, directly using each pair in a bi-temporal fusion and generating additional synthetic changes would cause the model to ignore unknown changes. Instead, we treat each image independently and generate synthetic changes by perturbing its own features.

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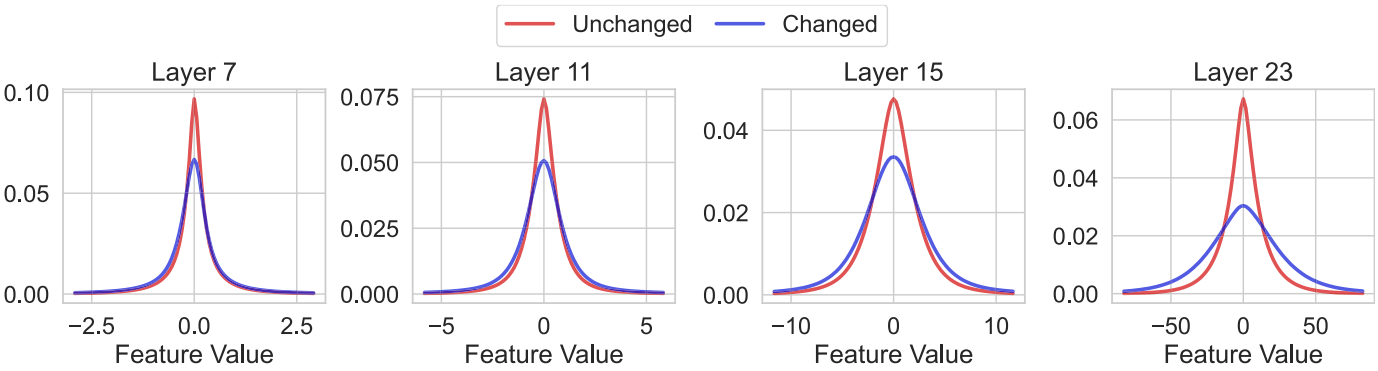


Fig. 3. Histogram plot of feature differences $f_1^{(l)} - f_2^{(l)}$, averaged across all five datasets and all channels per layer. Unchanged regions are narrowly concentrated near zero, while changed regions exhibit broader variation, especially in deeper layers. Both distributions can be approximated by a zero-centered Gaussian, but with different variance parameters. This directly motivates our latent-space change generation strategy.

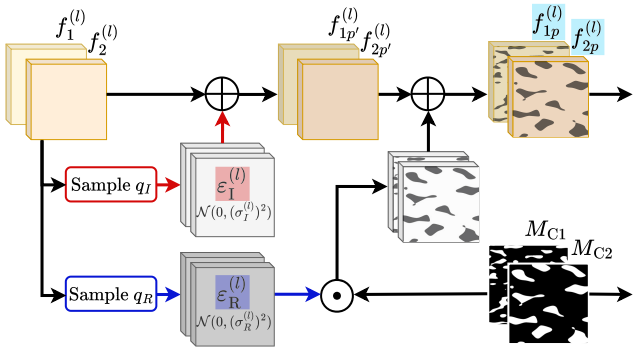


Fig. 4. Illustration of the change generation process.

We divide changes between images into two categories.

1) Irrelevant changes, such as illumination shifts, minor vegetation growth, or seasonal changes.
2) Relevant changes, which correspond to large changes like building construction or landslide damage.

Based on our analysis in Section III-B, we simulate these two types of changes using zero-centered Gaussian noise with a separate variance parameter, which is dynamically estimated to adapt to diverse scenes.

1) Irrelevant Change Noise: Empirically, the distribution of feature differences between unchanged regions follows a Gaussian-like shape (see Section III-B). Motivated by this and the maximum entropy principle (see Section III-B), we model irrelevant changes using Gaussian noise. Specifically, we sample a noise ε_{I}

$$\varepsilon_{\mathrm{I}}^{(l)} \sim \mathcal{N}\left(0, \left(\sigma_I^{(l)}\right)^2\right) \tag{2}$$

where the $\sigma_I^{(l)}$ is set to the q_Ith quantile of the absolute feature difference of specific feature layer l

$$\sigma_I^{(l)} = \text{Quantile}\left(\left|f_1^{(l)} - f_2^{(l)}\right|, q_I\right) \tag{3}$$

where q_I is a learnable parameter, which is initialized with a lower value (in our case, 0.85) to approximate the narrow distribution observed in our analysis. Quantile sampling is performed dynamically for each layer and feature channel within every training batch, allowing the noise magnitude to adapt to local feature statistics. The *key idea* is sampling from feature differences, which enables the extraction of irrelevant variation directly from the feature differences. This enables estimation of intraimage irrelevant variation without relying on fixed values.

Without injecting ε_{I}, the model would associate unchanged regions with exactly zero feature difference, making it overly sensitive to minor variations at test time.

2) Relevant Change Noise: Similarly, motivated by our feature analysis in Section III-B, the distribution of feature differences in changed regions approximately follows a Gaussian pattern. To approximate this, noise ε_{R} is sampled as

$$\varepsilon_{\mathrm{R}}^{(l)} \sim \mathcal{N}\left(0, \left(\sigma_R^{(l)}\right)^2\right) \tag{4}$$

where $\sigma_R^{(l)}$ is the q_Rth quantile computed from the concatenation of both feature sets along the batch dimension

$$\sigma_R^{(l)} = \text{Quantile}\left(\text{Concat}\left(\left[f_1^{(l)}, f_2^{(l)}\right], \text{dim} = \text{batch}\right), q_R\right). \tag{5}$$

We sample quantiles from both feature sets (concatenation) so the sampled value stays within the realistic bounds. q_R is a learnable parameter initialized to a value close to one (in our case, 0.98), so it captures the broader variance observed for relevant (changed) regions. Higher quantile initialization paired with a low learning rate also encourages that the irrelevant and relevant noises do not converge to the same value. As with irrelevant noise, the quantiles are sampled independently per channel each iteration.

Since the model operates in a binary setting, this variation captures all major changes, without explicitly differentiating the change class (i.e., building and landslide). We demonstrate extension to type-specific filtering at the end of Section IV-A.

The noise is applied selectively using a binary mask M_{C}, generated from thresholded Perlin noise [56] (see Fig. 4). The mask ensures that the changed region is random but spatially consistent, while conveniently serving as the ground truth during training.

Fig. 5 visually illustrates an example of real relevant (changed) and irrelevant (unchanged) variation from CLCD [57]. The bottom part of the figure compares feature differences between the real example and our synthetic approach,

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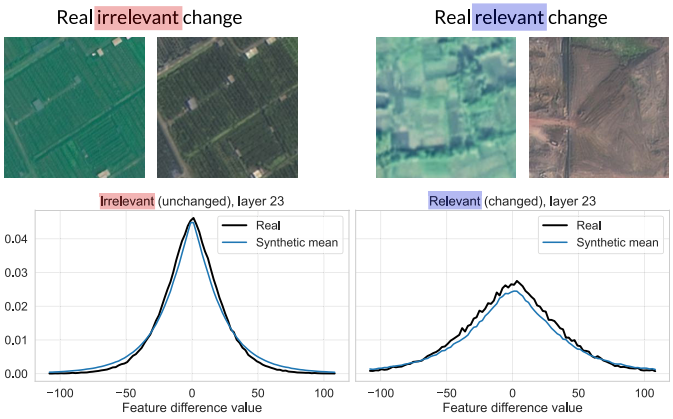


Fig. 5. (Top) Illustration of true irrelevant (unchanged) and relevant (changed) changes from CLCD [57]. (Bottom) Histograms comparing real and synthetic feature-difference distributions. Our approach approximates the true distribution of feature differences between changed and unchanged regions.

obtained by randomly sampling Gaussian noise using the above approach ten times. The simulated perturbations sufficiently approximate the true distributions of irrelevant and relevant variation for the purpose of training.

3) Final Synthetic Pairs: We generate the final synthetic training pairs (also depicted in Fig. 4) for each layer as

$$\left(f_1^{(l)}, f_1^{(l)} + \varepsilon_{\mathrm{I}}^{(l)} + M_{\mathrm{C1}} \odot \varepsilon_{\mathrm{R}}^{(l)}\right)\left(f_2^{(l)}, f_2^{(l)} + \varepsilon_{\mathrm{I}}^{(l)} + M_{\mathrm{C2}} \odot \varepsilon_{\mathrm{R}}^{(l)}\right) = \left(f_1^{(l)}, f_{1p}^{(l)}\right)\left(f_2^{(l)}, f_{2p}^{(l)}\right) \tag{6}$$

where $\odot$ denotes the elementwise multiplication. The synthetic training pairs are then assembled back into a set across all layers, resulting in two training pairs

$$\left(\left\{f_1^{(l)} | l \in L\right\}, \left\{f_{1p}^{(l)} | l \in L\right\}\right)\left(\left\{f_2^{(l)} | l \in L\right\}, \left\{f_{2p}^{(l)} | l \in L\right\}\right) = \left(F_1, F_1^p\right)\left(F_2, F_2^p\right) \tag{7}$$

where F_i^p denotes the synthetic perturbed feature set. This formulation enables the model to learn to detect relevant semantic changes while remaining robust to irrelevant variations, all without requiring any labeled change annotations.

Unlike prior works from other fields (e.g., anomaly detection) where noise is used as a static [47], [48], [52], nondecoupled [53] or robustness-oriented [46], [50] perturbation, MaSoN *dynamically* estimates two *decoupled* noise components (irrelevant and relevant) from feature statistics, a design motivated by the higher intraimage and interimage variability of remote sensing data.

D. Mask Decoder

To produce the final change mask prediction, feature pairs are first subtracted elementwise

$$\begin{aligned} F_i^{diff} &= F_i - F_i^p \quad \text{during training,} \quad \text{and} \\ F^{diff} &= F_1 - F_2 \quad \text{during inference.} \end{aligned} \tag{8}$$

The resulting set of feature differences is passed to the UPerNet decoder [58], which processes a hierarchy of feature differences to produce the final change map $M_{\mathrm{P}i}$, which is of the same spatial dimensions as the input image

$$M_{\mathrm{P}i} = \mathbb{D}\left(F_i^{diff}\right) . \tag{9}$$

Since we make two training pairs from a single input pair during training [see (6) and (7)], the decoder also predicts two maps M_{P1} and M_{P2}. During inference, the decoder produces a single map per feature pair (M_{P}, Fig. 2 right).

E. Training and Inference

The model is trained using Dice loss [59] where the prediction is the change map $M_{\mathrm{P}i}$ from (9) and the target is set to the binary mask $M_{\mathrm{C}i}$ from (6)

$$\mathcal{L} = \mathcal{L}_{Dice}\left(M_{\mathrm{P1}}, M_{\mathrm{C1}}\right) + \mathcal{L}_{Dice}\left(M_{\mathrm{P2}}, M_{\mathrm{C2}}\right) . \tag{10}$$

During inference, the change generation mechanism is removed, and the model directly predicts the change map from a pair of images (see Fig. 2 right). To obtain the binary mask, a sigmoid and a threshold of 0.5 are applied.

IV. Experiments

1) *Implementation Details:*
The following list summarises the implementation, evaluation, and dataset details used in our experiments.
As the encoder, we use the ViT-L [60] version of DINOv3 [61], which we do not finetune. Following related work [62], we take features from layers 7, 11, 15, and 23. Input images are cropped to 256 × 256 pixels. During training, we apply the flip and rotate augmentations. The model is trained using the AdamW [63] optimizer, with an initial learning rate of 10^{-5} for the decoder and 10^{-7} for the learnable quantile parameters (two values, one for q_I and one for q_R). A cosine scheduler without restarts is used. Each training run lasts 1000 iterations with a batch size of 16 and completes in approximately 7 min on a single NVIDIA A100 GPU. Additional details can be found in the Supplementary Section N.

2) *Evaluation Metrics and Datasets:*
Following standard practice [1], [2], [4], [22], [31], [64], we evaluate CD performance using **binary** precision, recall, and $F1$-score considering only **change class**. Experiments are repeated across five random seeds, and we report the mean and standard deviation. For each run, the model checkpoint from the final epoch is used. For training-free baselines, we use author-provided weights. To ensure robustness, we evaluate on five CD datasets covering *worldwide locations* and spanning *diverse change types*, from *building* and *urban* changes to *cropland* and *natural disaster* changes. The datasets cover different ground sample distances, sensors, and global geographic regions. *SYSU* [65] includes multiple change types such as buildings, groundwork, vegetation, and sea changes. *LEVIR* [30] focuses on building changes, while *CLCD* [57] captures cropland changes. We also include datasets that are particularly challenging for unsupervised methods: *GVLM* [66], containing natural disaster changes resulting from landslides, and *OSCD* [67], a low-resolution Sentinel-2 dataset covering worldwide urban changes. The models are trained on the dedicated training sets (using no labels) and evaluated on the corresponding test sets. Additional dataset details are provided in Supplementary Section B.

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TABLE I

UNSUPERVISED CD RESULTS ACROSS FIVE DIVERSE DATASETS AND THEIR AVERAGE. WE REPORT PRECISION (PR.), RECALL (RE.), AND $F1$ SCORE. RESULTS OF ADDITIONAL METRICS FOR INDIVIDUAL DATASETS ARE PROVIDED IN THE SUPPLEMENTARY SECTION E. FPS WAS BENCHMARKED ON AN NVIDIA A100 (DETAILS AND MORE COMPUTATION METRICS ARE IN SUPPLEMENTARY SECTION H). FIRST AND SECOND PLACE RESULTS ARE MARKED

	FPS [img/s]	Param. [M]	SYSU F1	LEVIR F1	GVLM F1	CLCD F1	OSCD F1	*Average* Pr.	*Average* Re.	*Average* F1
Pixel Difference [26]	456.4±54.5	0.0	42.9	8.9	23.4	16.1	18.5	16.2	49.3	22.0
DCVA [16]TGRS19	0.4±0.0	0.5	2.5	0.2	3.2	3.1	38.9	30.0	9.3	9.6
DINOv3-CVA [4]	24.4±0.3	303.1	56.5	19.1	19.3	27.1	21.3	19.0	**80.4**	28.7
AnyChange [4]NeurIPS24	0.4±0.0	641.1	46.0	25.4	20.8	26.5	26.1	26.0	43.0	29.0
Changen2-S9 [24]TPAMI25	33.3±0.1	99.9	44.8	19.9	19.1	15.6	1.5	17.3	39.4	20.2
I3PE [20]ISPRS23	65.9±0.2	24.9	33.7	16.8	20.3	11.5	3.6	15.6	24.8	17.2
HySCDG [19]CVPR25	41.0±0.1	65.1	52.7	15.4	17.3	24.9	17.2	29.2	65.2	25.5
SCM [42]IGARSS24	1.2±0.0	223.5	25.7	27.3	28.0	18.9	16.6	21.6	30.1	23.3
DynEarth [17]AAAI26	0.3±0.0	877.6	56.2	46.2	16.6	29.5	26.6	30.8	45.9	35.0
DynE. (DINOv3) [17]AAAI26	0.3±0.0	877.6	60.3	**49.5**	19.3	33.7	18.2	34.9	42.0	36.2
S2C (DINOv3) [23]AAAI26	3.8±0.0	303.6	37.6	32.0	38.1	52.2	5.1	41.4	42.1	36.5
MaSoN	39.7±0.1	337.5	**60.6** (±2.2)	38.9 (±1.2)	**55.4** (±2.9)	**54.3** (±1.5)	**43.5** (±1.0)	**44.4** (±1.9)	67.2 (±3.8)	**50.6** (±0.4)

A. Experimental Results

1) Unsupervised CD: We compare MaSoN against a range of unsupervised CD methods. We include image pixel differencing [26], [68], DCVA [16], and DINOv3 [61]-based change vector analysis (CVA) [4]. We include state-of-the-art methods HySCDG [19], I3PE [20], Changen2 [24], AnyChange [4], SCM [42], DynamicEarth [17], and S2C [23]. Their implementation details are provided in the Supplementary Section P. Results are summarized in Table I.

MaSoN achieves the highest average $F1$-score of 50.6, surpassing the previous best S2C [23] by 14.1 p.p. on average (a 45.41% relative gain). S2C, which trains by generating changes in pixel spaces using augmentations, performs well on CLCD; it degrades substantially in other settings. We attribute MaSoN's better performance to *data-grounded latent-space change synthesis*, which produces diverse, realistic semantic variations directly in a more robust latent representation space.

2) Comparison Against Training-Free Methods: AnyChange [4], SCM [42], and DynamicEarth [17], all built on SAM [18], underperform MaSoN on CLCD, GVLM, and OSCD. DynamicEarth exceeds MaSoN only on LEVIR (building CD) as it produces more precise building borders (see qualitative results discussion), consistent with SAM's strength in object-centric scenes. Overall, this suggests that training-free SAM does not always transfer well to remote sensing regimes that are less object-centric and/or lower resolution, where foreground–background cues are weaker than in natural images.

3) Comparison Against Pixel-Space Change Generation: Pixel-space synthesis is consistently less effective than latent-space generation. S2C [23] and I3PE [20] results suggest that hand-crafted pixel transformations cannot capture the breadth of possible changes. Even deep generative pixel-space methods (Changen2 [24] and HySCDG [19]) remain far behind MaSoN, supporting the view that pixel-level synthetic data generalizes poorly across diverse change types and acquisition conditions.

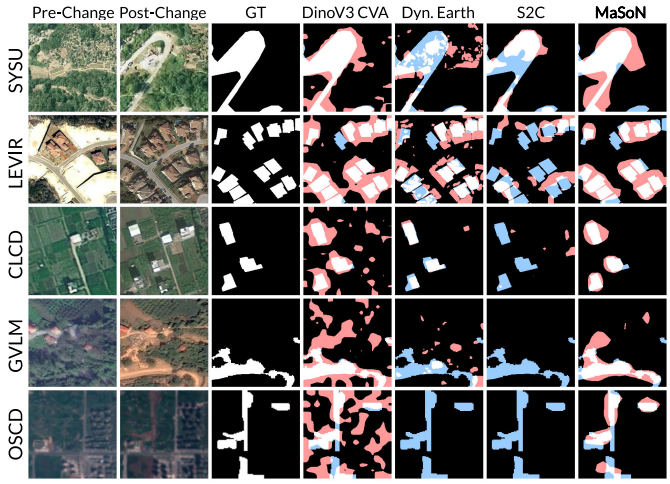


Fig. 6. Qualitative comparison of the predictions. The pair of considered images is shown in the first and second columns, followed by the ground-truth mask and predictions for each method. False positives are marked in red and false negatives in blue. Additional qualitative results, including discussion and failure cases, are presented in Supplementary Section G.

4) Comparison Against Traditional Methods: Compared to CVA [37] computed on features from the same DINOv3 [61] encoder, MaSoN doubles precision and improves average $F1$ by 21.9 p.p. This confirms that strong representations help, but MaSoN's gains are not due to the backbone alone. The learnable pipeline enabled by latent-space change synthesis is essential for turning features into accurate unsupervised CD.

5) Qualitative Examples: Fig. 6 illustrates MaSoN's strong generalization across diverse change types, particularly on GVLM (landslides) and OSCD (low-resolution), where previous SOTA methods DynamicEarth and S2C fail. On LEVIR, MaSoN occasionally overextends building regions, an issue shared by most methods except DynamicEarth, which uses SAM and is more biased toward object-centric detection. While DINOv3-based CVA captures most changes, it does so

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TABLE II
RESULTS ON THE OMBRIA SENTINEL-1 SAR FLOOD CHANGE-DETECTION DATASET

Method	Precision	Recall	F1
Pixel Diff.	35.96	43.33	39.98
Cop.-FM CVA	46.59	47.65	47.11
MaSoN	38.13 ±0.8	91.87 ±1.3	53.88 ±0.6

at the cost of excessive false positives, an issue that MaSoN substantially reduces.

6) Extension to Specific Changes Filtering: Certain applications may require isolating or filtering specific types of changes. To address this, MaSoN can be extended with a text-guided filtering procedure similar to one from DynamicEarth [17] (see Supplementary Section R). To evaluate its performance in this scenario, we run the experiment on the LEVIR [30] dataset, strictly focusing on buildings. The performance *improves from* 38.9% $F1$ *to* 50.3% $F1$, surpassing the original DynamicEarth, which achieves 49.5%. As seen in Table I, MaSoN already outperforms DynamicEarth on all other datasets without filtering. We experimented with filtering across other datasets; however, this did not yield consistent improvements. We hypothesize that this comes from the complex nature of many change types (e.g., landslides or cropland), which are difficult to describe unambiguously by text. Consequently, we evaluated filtering on LEVIR, as building changes are textually well-defined. These findings highlight both MaSoN's flexibility and the current limitation of text-based filtering for domain-specific changes.

7) Extension to a New Modality: The RGB modality can be unreliable in certain UCD scenarios, for example, for cloud-covered flood imagery. MaSoN can be directly applied to SAR data by replacing the RGB encoder with an SAR-capable alternative, such as Copernicus-FM [69]. On the Sentinel-1 OMBRIA flood-change dataset [70] (see Table II), MaSoN outperforms both pixel differencing and CVA when using Copernicus-FM features. Other methods in Table I cannot be easily extended to SAR, as they rely on RGB-only SAM or restrict their change-generation mechanisms to the RGB domain.

We also extend MaSoN to the multispectral Sentinel-2 dataset using the DEO [71] backbone and evaluate it on OSCD-multispectral. It achieves an $F1$-score of 45.1, which is better than RGB-MaSoN and all related methods that process only the RGB part of OSCD (see Table I).

We further evaluate MaSoN on heterogeneous optical–SAR CD. We use co-registered Landsat-8 optical and Sentinel-1A SAR observations of flooding in California from [72] and process both with Copernicus-FM [69] to obtain features. MaSoN obtains 22.56 $F1$, compared with 12.84 $F1$ for feature CVA using the same Copernicus-FM encoder. Thus, the learned latent-perturbation framework provides a clear improvement over direct feature comparison, but the absolute heterogeneous performance remains limited compared to rule-based methods like RIEM [41], which achieves 49.9 $F1$.

V. ABLATION STUDY

In this section, we experimentally validate the contributions proposed in this article. We report the average $F1$ of five datasets repeated with five random seeds for every experiment. The full individual dataset results with additional metrics are in Supplementary Section G, additional ablations in Supplementary Section K, and ablation implementation details in Supplementary Section R.

TABLE III
CHANGE GENERATION ABLATION

ID		*Avg.* F1
Ours	Fully latent generation	50.6±0.4
Rect. M_C	Random rectangles M_C	46.9±0.8
No M_C	No bin. mask for changed	36.5±1.7
CycGan	Irrelevant with CycGAN	36.9±0.7
No Dyn.	No dynamic estimation	25.3±1.9
No Irrel.	No irrelevant changes	15.6±0.8
Pix. Gen.	Pixel noise generation	12.8±1.8

A. Core Ablation Studies

We validate our latent-space change generation strategy through a series of ablations presented in Table III. Replacing the binarised Perlin mask with a random rectangle mask (**Rect.** M_C) slightly lowers performance because most changes are nonrectangular, and since our approach models them better due to its higher randomness. Removing the binary mask for relevant noise (**No** M_C), where noise is instead applied to half of the features and the ground truth for those is set to "all changed," reduces performance by 14.1 p.p., confirming the benefit of spatially structured perturbations.

Next, we replace the irrelevant latent noise ε_I with CycleGAN [73]-transformed images that simulate irrelevant changes in pixel space via a learned transformation (**CycGan**). This reduces the $F1$ by 13.7 p.p., highlighting the advantage of modeling irrelevant variation directly in latent space using our approach.

Removing dynamic noise estimation (as described in Section III-C) and instead using a fixed noise magnitude reduces performance by 25.3 p.p. (**No Dyn.**). This demonstrates the importance of data-driven estimation of noise from features that change during training and vary across geographical domains. Eliminating irrelevant change noise [**No Irrel.**, refer to (2)] decreases the performance by 35.0 p.p. as the model incorrectly learns that unchanged regions correspond to zero feature difference.

Together, these results show that both mechanisms are essential. Unlike machine learning works from other fields (e.g., anomaly detection with low intrasample variation, see Section II), MaSoN *dynamically* estimates two *decoupled* noise components based on feature statistics. In contrast to these other fields, *we show that this is crucial* for bi-temporal remote sensing imagery, where variability is high across geographical regions and within image pairs themselves.

Applying the noise perturbation directly to the pixels of input images (**Pix. Gen.**) leads to the poorest result (12.8 $F1$), showing that raw pixel-level noise lacks the semantic robustness and is not as nicely behaved as latent feature space.

B. Quantile Sampling Dimension

MaSoN calculates the quantiles per channel in batch (see Section III-C). It could, however, use the same value across all

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TABLE IV
NOISE SAMPLING DIMENSION

	Avg. F1
Per-channel in batch (Ours)	50.6±0.4
Per-channel in sample	29.6±10.7
Per-sample	44.3±0.8
Per-batch	32.9±1.7

TABLE V
BACKBONE ABLATION

Architecture	*Avg.* F1
ViT-L (DINOv3) (Ours)	50.0±0.7
ViT-L (DINOv2)	47.4±1.5
ViT-B (DINOv3)	45.4±2.3
Swin-T (ADE20k)	41.7±1.2
ResNet50 (ADE20k)	34.1±0.5

TABLE VI
TARGET LAYERS FOR NOISE

L. 7	L. 11	L. 15	L. 23	*Avg.* F1
✓	✓	✓	✓	50.6±0.4
✓	✓	✓		48.6±0.3
✓	✓		✓	44.6±1.6
✓		✓	✓	46.2±1.6
	✓	✓	✓	48.7±1.2

channels in a batch, or separate values for each sample, or even separate values for each channel within a sample. The results of such strategies are presented in Table IV. We hypothesize that our strategy of calculating statistics per channel in batch yields the best performance due to generalization, since the scope of sampling is neither too general (per-batch) nor too specific (per-channel).

C. Backbone Ablation

MaSoN's performance does not stem solely from DINOv3 [61], but it does achieve better performance with a stronger encoder, as shown in Table V. If we replace it with DINOv2 [74] and some weaker backbones like Swin [75] and ResNet50 [76], it still remains effective and performs better or comparable to existing approaches. We also opt for DINOv3 as it is also used by two of the best-performing competing methods.

D. Layer Noise Importance

To verify the importance of injecting noise into each layer, we conducted an experiment where we excluded noise injection for individual layers. The results in Table VI show that not applying noise to the middle 15th layer brings the largest drop in performance (6 p.p.). Nevertheless, all four layers play an important role in the overall performance.

E. Feature Analysis

We analyze how the changes, synthesized by adding noise in the latent space, are expressed when reconstructed to pixel space. We train a simple UNet [77] decoder to reconstruct the input image from the features (see Supplementary Section E for more details). The results shown in Fig. 7 display both the reconstructions without noise and with the addition of either irrelevant or relevant noise, where noise is generated as explained in Section III-C. The changes synthesized using our methodology in the latent space also represent a meaningful shift in the pixel space. This further demonstrates how a well-behaved structure of latent space enables a mechanism as simple as Gaussian noise to produce changes that approximate real changes. Based on this and the insights from the analysis in Section III-B, we believe that latent-space generation has significant potential for future work.

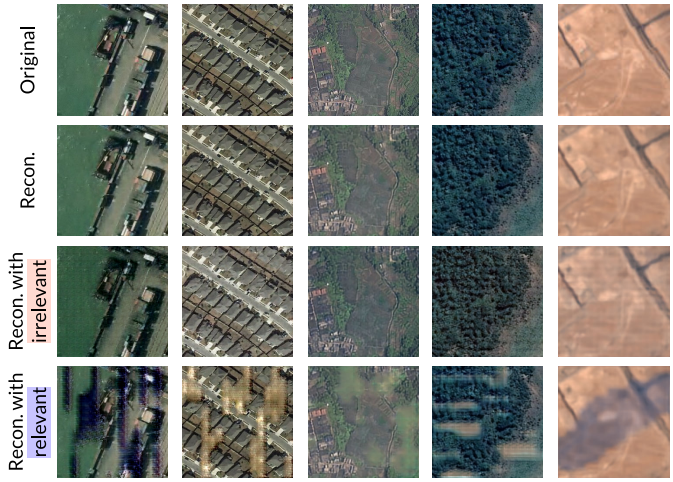


Fig. 7. First row contains original input image, second row reconstructions from features, third row reconstructions with irrelevant noise added to features, and the final row reconstructions with relevant noise added to selected regions (as explained in Section III-C). The addition of smaller irrelevant noise in the latent space is reflected in a minor color and texture variation. The addition of relevant noise is expressed as a significant but coherent change, even in reconstructed pixel space.

We see how the feature difference L2 map is expressed for real changes and synthetic changes in Fig. 8. The range of synthetic changes approximates the one in real ones, while the spatial distribution of change is different due to randomness and since each image in a pair is treated as a single temporal (see Section III-C).

F. Limitations and Future Work

MaSoN could be viewed as a self-supervised approach within the broader field of unsupervised CD [3]. Consequently, its limitation is the need for light finetuning rather than full training-free operation. Given its stronger accuracy than zero-shot alternatives and its short training time, we consider this a reasonable tradeoff. Moreover, even compared to current zero-shot methods that are prohibitively slow at inference (e.g., DynamicEarth processes only 0.4 images/s), MaSoN is more time-efficient (including fine-tuning) for data collections larger than roughly 170 images. We believe an important next step for unsupervised methods is to reduce end-to-end latency by accelerating latent change synthesis and to improve inference throughput. Another limitation is the heterogeneous CD and very subtle changes, where MaSoN explicitly relies on

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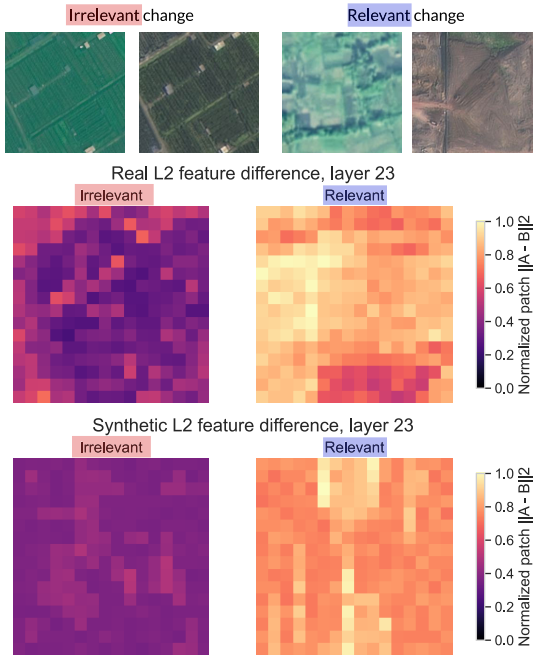


Fig. 8. First row contains visual examples being considered, second row shows real feature difference L2 map, and the last row illustrates the feature difference L2 map with synthetically generated changes using our approach. Our strategy approximates the true change range, while the spatial distribution is more random (see Section III-C).

alignment and discriminability of features in encoders' latent space, which is not the case in some complex scenarios. Cross-modal alignment of the encoder thus remains an important direction of future work. Additional discussion and societal impact are provided in Supplementary Section A.

VI. Conclusion

We propose MaSoN, a framework for unsupervised CD (UCD) that addresses the limited generalization of existing approaches, which rely on priors from foundation models in a training-free manner or train on changes generated in pixel space. MaSoN trains on synthetic changes generated via Gaussian noise injected directly into the latent feature space, where the noise is dynamically estimated per dataset using the data's own statistics. This enables highly diverse, data-driven change generation without external data or auxiliary generative models. MaSoN yields substantial improvements, outperforming prior unsupervised state of the art by 14.1 p.p. on average across five diverse benchmarks. MaSoN also extends to multispectral and SAR modalities via a simple encoder swap, achieving great results. Beyond UCD, this work establishes latent-space synthesis as a viable and effective direction for unsupervised learning in general. The idea holds promise for sample generation and representation learning in low-data regimes, offering a path toward more general and data-efficient vision models in remote sensing as well as other domains.

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